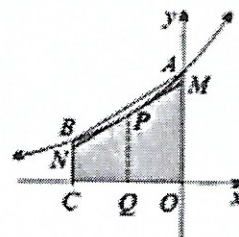


Inequalities WS 1

- Suppose that a and b are real numbers. Prove that $\frac{a^2+b^2}{2} \geq \left(\frac{a+b}{2}\right)^2$
- If $x, y > 0$, prove that $\frac{1}{2}(x^3 + y^3) \geq \left(\frac{x+y}{2}\right)^3$
- Show that $(a^2 - b^2)(c^2 - d^2) \leq (ac - bd)^2$. Deduce that $(a^2 - b^2)(a^4 - b^4) \leq (a^3 - b^3)^2$
- If $t > 0$ show that $\frac{1}{(1+t)^2} < \frac{1}{1+t} < 1$. By integrating between 0 and u , deduce that for $u > 0$, $\frac{u}{1+u} < \ln(1+u) < u$.
- Show that $(lx + my + nz)^2 \leq (l^2 + m^2 + n^2)(x^2 + y^2 + z^2)$. Deduce that
 - $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$
 - $(a^3 + b^3 + c^3)^2 \leq (a^2 + b^2 + c^2)(a^4 + b^4 + c^4)$
- Prove that for $a, b, c, d \geq 0$, $\frac{a+b+c+d}{4} \geq (abcd)^{\frac{1}{4}}$
- If $0 < t < 1$ show that $\frac{1}{2} < \frac{1}{1+t} < 1$. By integrating between 0 and u , deduce that for $0 < u < 1$, $\frac{u}{2} < \ln(1+u) < u$.
- Consider the function $f(x) = (ax - b)^2 + (cx - d)^2$ where a, b, c and $d \in \mathbb{R}$.
 - Explain why $f(x) \geq 0$ for all x
 - Hence or otherwise, prove that $|ab + cd| \leq \sqrt{a^2 + c^2}\sqrt{b^2 + d^2}$
- The diagram shows the points $A(0,1)$ and $B(-1, e^{-1})$ on the curve $y = e^x$, and the point $C(-1,0)$ on the x -axis. The tangent at $P\left(-\frac{1}{2}, e^{-\frac{1}{2}}\right)$ intersects OA at M and BC at N .
 - Determine the exact area of the region bounded by the curve, BC , CO and OA .
 - Find the area of the trapezium $OABC$,
 - Show that the area of $OMNC$ is equal to $CO \times PQ$ and hence find the area of the trapezium.
 - Hence show that $\frac{1}{2}(3 + \sqrt{5}) < e < 3$.
- If $0 < a < b$ show that $a < \frac{a+b}{2} < b$
 - If $0 < a < b$ show that $a < \sqrt{ab} < b$
- If $f(x)$, $g(x)$ and $h(x)$ are distinct non-negative continuous functions of x in the interval $a \leq x \leq b$ and $f(x) < g(x) < h(x)$, explain why $\int_a^b f(x)dx < \int_a^b g(x)dx < \int_a^b h(x)dx$
 - By considering the interval $0 < x < 1$ as an inequality, use algebra to show that $\frac{1}{2}x(1-x)^3 < \frac{x(1-x)^3}{1+x} < x(1-x)^3$
 - Deduce that $\frac{1}{2} \int_0^1 x(1-x)^3 dx < \int_0^1 \frac{x(1-x)^3}{1+x} dx < \int_0^1 x(1-x)^3 dx$
 - Given that $\int_0^1 \left(\frac{x(1-x)^3}{1+x}\right) dx = \frac{67}{12} - 8 \ln 2$, deduce that $\frac{83}{120} < \ln 2 < \frac{667}{960}$
- Show that for $x > 0$, $x - \frac{1}{2}x^2 < \ln(1+x) < x - \frac{1}{2}x^2 + \frac{1}{3}x^3$
 - Show that for $x > 0$, $x - \frac{1}{3}x^3 < \tan^{-1} x < x - \frac{1}{3}x^3 + \frac{1}{5}x^5$



Answer

8. a. $(1 - e^{-1})$ sq. unit; b. $\frac{1}{2}(1 + e^{-1})$ sq. units; c. $e^{-\frac{1}{2}}$ sq. units